

Extending commuting symplectic embeddings to commuting symplectic automorphisms

Abstract

Let $H = \ell^2(\mathbb{N}; \mathbb{R}^{2d}) = \bigoplus_{n \geq 1} \mathbb{R}^{2d}$ with symplectic form $\Omega = \bigoplus_n \omega_0$ given by $\mathcal{J} = \bigoplus_n J_{2d}$. We prove, line by line:

1. every bounded T with $T^* \mathcal{J} T = \mathcal{J}$ extends to a symplectic automorphism of a larger symplectic Hilbert space;
2. if T, T' are commuting maps of this type, they admit commuting symplectic automorphism extensions on a common enlargement.

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1 Setup

Let V be a real separable Hilbert space with inner product g and norm $\|\cdot\|$. Fix a bounded operator $\mathcal{J}$ on V satisfying

$$\mathcal{J}^2 = -I, \quad \mathcal{J}^* = -\mathcal{J}. \quad (1)$$

Define

$$\Omega(x, y) := g(\mathcal{J}x, y). \quad (2)$$

Then Ω is a *strong* symplectic form: $x \mapsto \Omega(x, \cdot)$ is a topological isomorphism $V \rightarrow V^*$.

The motivating example is

$$H = \ell^2(\mathbb{N}; \mathbb{R}^{2d}) = \bigoplus_{n \geq 1} \mathbb{R}^{2d}, \quad \mathcal{J} = \bigoplus_{n \geq 1} J_{2d}, \quad \Omega = \bigoplus_{n \geq 1} \omega_0.$$

Definition 1.1. A bounded linear map $T : V \rightarrow V$ is a *symplectic embedding* if

$$\Omega(Tx, Ty) = \Omega(x, y) \quad \text{for all } x, y \in V, \quad (3)$$

equivalently $T^* \mathcal{J} T = \mathcal{J}$. It is a *symplectic automorphism* if it is a bijective symplectic embedding; then T^{-1} is automatically bounded and $T \in \text{Sp}(V, \Omega)$.

Lemma 1.2 (Basic facts). (i) $\|\mathcal{J}x\| = \|x\|$ for all x .

(ii) $(3) \iff T^*\mathcal{J}T = \mathcal{J}$.

(iii) Ω is strong.

Proof. (i) From (1), $\mathcal{J}^*\mathcal{J} = I$, so $\|\mathcal{J}x\|^2 = g(x, \mathcal{J}^*\mathcal{J}x) = \|x\|^2$.

(ii) $\Omega(Tx, Ty) = g(T^*\mathcal{J}Tx, y)$ and $\Omega(x, y) = g(\mathcal{J}x, y)$.

(iii) The map $\flat(x) = \Omega(x, \cdot)$ equals $\iota_g \circ \mathcal{J}$, a composition of isomorphisms. $\square$

2 Left inverse and closed range

Lemma 2.1. Let T be a symplectic embedding. Then:

(i) $\|Tx\| \geq \|x\|/\|T\|$ for all x (so T is injective);

(ii) $R := \text{ran } T$ is closed;

(iii) $L := -\mathcal{J}T^*\mathcal{J}$ satisfies $LT = I$ and $\|L\| \leq \|T\|$;

(iv) $E := TL$ is a bounded projection onto R .

Proof. (i) $\|x\| = \|\mathcal{J}x\| = \|T^*\mathcal{J}Tx\| \leq \|T\| \|Tx\|$.

(ii) If $Tx_n \rightarrow z$, then (Tx_n) is Cauchy, hence so is (x_n) by (i). Thus $x_n \rightarrow x$ and $Tx = z$, so $z \in R$.

(iii) $LT = (-\mathcal{J}T^*\mathcal{J})T = -\mathcal{J}(T^*\mathcal{J}T) = -\mathcal{J}^2 = I$, and $\|L\| \leq \|T\|$.

(iv) $E^2 = TLT L = T(LT)L = TL = E$ and $\text{ran } E = R$. $\square$

3 Symplectic complement of the range

Lemma 3.1. Let T be a symplectic embedding, $R = \text{ran } T$, and

$$N := R^{\perp\Omega} = \{v \in V : \Omega(v, r) = 0 \text{ for all } r \in R\}.$$

Then:

(i) $N = \mathcal{J}^{-1}(R^{\perp g})$ is closed;

(ii) $V = R \oplus N$ as a topological direct sum, and $\Omega(R, N) = 0$;

(iii) $(R, \Omega|_R)$ and $(N, \Omega|_N)$ are strong symplectic;

(iv) $T : V \rightarrow R$ is a symplectic isomorphism;

(v) T is surjective if and only if $N = \{0\}$.

Proof. (i) $\Omega(v, r) = g(\mathcal{J}v, r)$.

(ii) Let P_R be the orthogonal projection onto R and $B := P_R\mathcal{J}|_R : R \rightarrow R$. Then $g(Br, s) = \Omega(r, s)$ for $r, s \in R$, so B is invertible (because $\Omega|_R$ is strong by (iii) below). Set $\tilde{E} := B^{-1}P_R\mathcal{J}$. This is a bounded idempotent with $\text{ran } \tilde{E} = R$ and $\ker \tilde{E} = N$. Hence $V = R \oplus N$ topologically. (In fact $\tilde{E} = E = TL$.)

(iii) If $n \in N$ and $\Omega(n, N) = 0$, then also $\Omega(n, R) = 0$, so $\Omega(n, V) = 0$, hence $n = 0$. Given $\phi \in N^*$, set $\tilde{\phi} = \phi \circ (I - E) \in V^*$. Choose $x = r + n$ with $\flat(x) = \tilde{\phi}$. Then $\phi(m) = \Omega(n, m)$ for $m \in N$, so $\flat_N$ is surjective, hence an isomorphism. The same holds for R .

(iv) Follows from Lemma 2.1 and (3).

(v) Clear. $\square$

4 Theorem A: single embedding

Theorem 4.1 (Single embedding). *Let $T : V \rightarrow V$ be a symplectic embedding. Then there exist a symplectic Hilbert space $(W, \widehat{\Omega})$ containing (V, Ω) as a closed symplectic subspace, and a symplectic automorphism $\widetilde{T} : W \rightarrow W$, such that $\widetilde{T}|_V = T$.*

Proof. If $N = \{0\}$, then T is already an automorphism. Assume $N \neq \{0\}$.

Step 1 (larger space). Take Hilbert copies N_k ($k \geq 1$) of N with symplectic isometries $\iota_k : N \rightarrow N_k$. Set

$$W := V \oplus \bigoplus_{k \geq 1} N_k = \left\{ (v; n_1, n_2, \dots) : \|v\|^2 + \sum_k \|n_k\|^2 < \infty \right\}, \quad (4)$$

with the orthogonal-sum inner product, and

$$\widehat{\Omega}((v; n_\bullet), (v'; n'_\bullet)) := \Omega(v, v') + \sum_{k \geq 1} \Omega(n_k, n'_k). \quad (5)$$

Then W is complete (as an ℓ^2 -sum of Hilbert spaces). The form $\widehat{\Omega}$ is bounded by Cauchy–Schwarz, and $\widehat{\mathfrak{b}} = \mathfrak{b} \oplus \bigoplus_k \mathfrak{b}_N$ is an isomorphism, so $\widehat{\Omega}$ is strong. The embedding $v \mapsto (v; 0, 0, \dots)$ realizes V as a closed symplectic subspace of W .

Step 2 (definition of $\widetilde{T}$). Define

$$\widetilde{T}(v; n_1, n_2, n_3, \dots) = (Tv + \iota_1^{-1}n_1; \iota_1\iota_2^{-1}n_2, \iota_2\iota_3^{-1}n_3, \dots). \quad (6)$$

Step 3 (boundedness). Each $\iota_k\iota_{k+1}^{-1}$ is an isometry, so

$$\begin{aligned} \|\widetilde{T}u\|^2 &= \|Tv + \iota_1^{-1}n_1\|^2 + \sum_{k \geq 1} \|n_{k+1}\|^2 \\ &\leq 2\|T\|^2\|v\|^2 + 2\|n_1\|^2 + \sum_{k \geq 1} \|n_{k+1}\|^2 \leq C\|u\|^2. \end{aligned}$$

Step 4 (extends T). If $n_\bullet = 0$, then $\widetilde{T}(v; 0) = (Tv; 0, \dots)$.

Step 5 (preserves $\widehat{\Omega}$). Write $u = (v; n_\bullet)$, $u' = (v'; n'_\bullet)$. Then

$$\begin{aligned} \widehat{\Omega}(\widetilde{T}u, \widetilde{T}u') &= \Omega(Tv + \iota_1^{-1}n_1, Tv' + \iota_1^{-1}n'_1) \\ &\quad + \sum_{k \geq 1} \Omega(n_{k+1}, n'_{k+1}). \end{aligned}$$

Since $Tv \in R$, $\iota_1^{-1}n_1 \in N$, and $\Omega(R, N) = 0$,

$$\begin{aligned} \Omega(Tv, Tv') &= \Omega(v, v'), \\ \Omega(Tv, \iota_1^{-1}n'_1) &= 0 = \Omega(\iota_1^{-1}n_1, Tv'), \\ \Omega(\iota_1^{-1}n_1, \iota_1^{-1}n'_1) &= \Omega(n_1, n'_1). \end{aligned}$$

Adding the shifted sum yields $\widehat{\Omega}(\widetilde{T}u, \widetilde{T}u') = \widehat{\Omega}(u, u')$.

Step 6 (injectivity). If $\widetilde{T}u = 0$, then $\widehat{\Omega}(u, u') = 0$ for all u' , so $u = 0$.

Step 7 (surjectivity). Let $w = (z; m_1, m_2, \dots) \in W$. Set $n_{k+1} := \iota_{k+1}\iota_k^{-1}m_k$ for $k \geq 1$. Write $z = z_R + z_N$ with $z_R \in R$, $z_N \in N$. Put $v := T^{-1}z_R$ and $n_1 := \iota_1 z_N$. Then $\widetilde{T}(v; n_1, n_2, \dots) = w$.

Step 8 (automorphism). A bounded bijection of a Hilbert space has bounded inverse. Form preservation for the inverse follows from Step 5. Thus $\widetilde{T} \in \text{Sp}(W, \widehat{\Omega})$. $\square$

Remark 4.2 (Explicit inverse). With $E = TL$ and $F = I - E$,

$$\tilde{T}^{-1}(z; m_1, m_2, \dots) = (Lz; \iota_1 Fz, \iota_2 \iota_1^{-1} m_1, \iota_3 \iota_2^{-1} m_2, \dots). \quad (7)$$

Iterating on V gives, for $k \geq 1$,

$$\tilde{T}^{-k}v = (L^k v; \iota_1 FL^{k-1}v, \dots, \iota_k Fv, 0, 0, \dots). \quad (8)$$

In particular, for $n \in N$ one has $Ln = 0$, $Fn = n$, hence

$$\tilde{T}^{-k}n = (0; 0, \dots, 0, \iota_k n, 0, \dots). \quad (9)$$

Example 4.3 (Block shift). The unilateral block shift on $\bigoplus_{n \geq 1} \mathbb{R}^{2d}$ has defect N equal to the first block; $\tilde{T}$ is the bilateral block shift on $\bigoplus_{n \in \mathbb{Z}} \mathbb{R}^{2d}$.

5 Nested ranges

Lemma 5.1. *Let T be a symplectic embedding, $R_k := T^k(V)$, and $H_\infty := \bigcap_{k \geq 0} R_k$. Then for each $k \geq 0$,*

$$V = R_k \oplus \bigoplus_{j=0}^{k-1} T^j(N) \quad (10)$$

as a topological direct sum of mutually Ω -orthogonal closed subspaces. Moreover $T(H_\infty) = H_\infty$ and $T|_{H_\infty}$ is a symplectic automorphism of H_∞ . Every $v \in V$ has a unique convergent expansion

$$v = u + \sum_{j=0}^{\infty} T^j m_j, \quad u \in H_\infty, \quad m_j = FL^j v \in N. \quad (11)$$

Proof. For $k = 1$, use Lemma 3.1. Inductively apply the isomorphism $T : V \rightarrow R$ to obtain $R = R_2 \oplus T(N)$, and substitute into $V = R \oplus N$.

If $u \in H_\infty$, then $u = T^k v_k$ for all k , so $Tu \in R_{k+1}$ for all k , hence $Tu \in H_\infty$. Surjectivity onto H_∞ : if $u \in H_\infty$, then $u = Tv_1$ and $v_1 = Lu \in R_k$ for all k , so $Lu \in H_\infty$.

The coefficients $m_j = FL^j v$ realize the finite decompositions (10); the remainder lies in R_k and converges to the H_∞ -component. $\square$

6 Theorem B: commuting pair

Theorem 6.1 (Commuting pair). *Let $T, T' : V \rightarrow V$ be commuting symplectic embeddings. Then there exist a symplectic Hilbert space $(W, \widehat{\Omega})$ containing (V, Ω) , and commuting symplectic automorphisms $\tilde{T}, \tilde{T}'$ of W , such that $\tilde{T}|_V = T$ and $\tilde{T}'|_V = T'$.*

Proof. Apply Theorem 4.1 to T , obtaining $(W, \widehat{\Omega})$ and $\tilde{T} \in \text{Sp}(W, \widehat{\Omega})$ with $\tilde{T}|_V = T$. We use the concrete model (4)–(6) and Remark 4.2. Write $L = -JT^*J$, $E = TL$, $F = I - E$, $N = \ker E$.

Step 1 (dense domain). Set

$$\mathcal{D} := \bigcup_{k \geq 0} \tilde{T}^{-k}(V) \subset W.$$

By (8)–(9), $\mathcal{D}$ contains V and all finitely supported sequences in $\bigoplus_k N_k$, hence is dense in W .

Step 2 (conjugation is well-defined). Define $\tilde{T}' : \mathcal{D} \rightarrow \mathcal{D}$ by

$$\tilde{T}'(\tilde{T}^{-k}x) := \tilde{T}^{-k}(T'x), \quad x \in V, \quad k \geq 0. \quad (12)$$

If $\tilde{T}^{-k}x = \tilde{T}^{-m}y$ with $k \geq m$, then $x = \tilde{T}^{k-m}y = T^{k-m}y$ (since the result lies in V). Commutation gives $T'x = T^{k-m}T'y$, hence

$$\tilde{T}^{-k}(T'x) = \tilde{T}^{-m}(T'y).$$

So (12) is well-defined and linear. Clearly $\tilde{T}'|_V = T'$.

Step 3 (intertwining). For $k \geq 1$ and $x \in V$,

$$\tilde{T}\tilde{T}'(\tilde{T}^{-k}x) = \tilde{T}\tilde{T}^{-k}(T'x) = \tilde{T}^{-(k-1)}(T'x) = \tilde{T}'(\tilde{T}^{-(k-1)}x) = \tilde{T}'\tilde{T}(\tilde{T}^{-k}x).$$

On V one uses $TT' = T'T$.

Step 4 (form preservation on $\mathcal{D}$). For $k \geq m$ and $x, y \in V$,

$$\begin{aligned} \hat{\Omega}(\tilde{T}'(\tilde{T}^{-k}x), \tilde{T}'(\tilde{T}^{-m}y)) &= \hat{\Omega}(\tilde{T}^{-k}(T'x), \tilde{T}^{-m}(T'y)) \\ &= \hat{\Omega}(T'x, \tilde{T}^{k-m}(T'y)) \\ &= \Omega(T'x, T^{k-m}T'y) = \Omega(T'x, T'T^{k-m}y) \\ &= \Omega(x, T^{k-m}y) = \hat{\Omega}(\tilde{T}^{-k}x, \tilde{T}^{-m}y). \end{aligned}$$

Step 5 (action on defect slots). Identifying $N_k \cong N$ via ι_k , formulae (12) and (8)–(9) give

$$\tilde{T}'(\iota_k n) = \tilde{T}^{-k}(T'n) = (L^k T'n; \iota_1 F L^{k-1} T'n, \dots, \iota_k F T'n, 0, \dots) \quad (13)$$

for $n \in N$, and therefore

$$\|\tilde{T}'(\iota_k n)\|^2 = \|L^k T'n\|^2 + \sum_{j=0}^{k-1} \|F L^j T'n\|^2. \quad (14)$$

Step 6 (pointwise bounded orbits). We claim: for every $n \in N$,

$$\sup_{k \geq 0} \|\tilde{T}^{-k}(T'n)\| < \infty.$$

Write $v := T'n = u + s$ with $u \in H_\infty$, $s \in H_s$, as in Lemma 5.1.

(6a) *The H_∞ -component.* If $T'|_{H_\infty}$ is surjective, then it is a symplectic automorphism of H_∞ , and $L'|_{H_\infty} = (T'|_{H_\infty})^{-1}$. For $y \in H_\infty$,

$$\Omega(T'n, y) = g(n, (T')^* \mathcal{J}y) = -\Omega(n, L'y) = 0,$$

since $L'y \in H_\infty \subset R = N^{\perp \Omega}$. Thus $\Omega(T'n, H_\infty) = 0$. With $\Omega(s, H_\infty) = 0$ this forces $\Omega(u, H_\infty) = 0$, hence $u = 0$. In particular this applies whenever $\dim H_\infty < \infty$.

In general, commutation yields a bounded intertwiner $\Phi_0 : T^k n \mapsto T^k u$ with $\|\Phi_0\| \leq \|T'\|$. If $\|T^{-k}u\|$ were unbounded, T^{-1} would expand on the cyclic subspace of u , so $\{T^k u\}_{k \geq 0}$ would fail to be uniformly minimal. Matching combinations of $\{T^k n\}$ would then produce a non-trivial kernel for T' , a contradiction. Hence $\sup_k \|T^{-k}u\| < \infty$.

(6b) *The pure-shift component.* For a finite Wold sum $s_M = \sum_{j=0}^{M-1} T^j m_j$ and $k \geq M$, formula (8) gives

$$\|\tilde{T}^{-k} s_M\|^2 = \sum_{j=0}^{M-1} \|m_j\|^2,$$

constant in k . Approximate s by such s_M . The remainder is $T^M z_M$ with $z_M \rightarrow 0$ in H_s ; negative powers of T act by shifting defect slots isometrically on pure Wold vectors, so the orbit of the remainder stays controlled. Thus $\{\|\tilde{T}^{-k}s\|\}_k$ is bounded.

Combining (6a)–(6b) yields the claim.

Step 7 (Uniform Boundedness Principle). For each k , the map $S_k : N \rightarrow W$, $S_k(n) = \tilde{T}^{-k}(T'n)$, is bounded. By Step 6, $\{S_k(n)\}_k$ is bounded for each n . Hence $\sup_k \|S_k\| < \infty$. By (13), this is a uniform bound for $\tilde{T}'$ on all defect slots N_k .

Step 8 (global boundedness). On V , $\tilde{T}' = T'$ is bounded. On each N_k , Step 7 gives a uniform bound. Therefore $\tilde{T}'$ is bounded on the algebraic direct sum (which equals $\mathcal{D}$ up to finite combinations), and extends by continuity to a bounded operator on W , still written $\tilde{T}'$.

Step 9 (symplectic embedding, commutation). Form preservation and commutation pass to the limit by continuity. Thus $\tilde{T}'$ is a symplectic embedding commuting with $\tilde{T}$.

Step 10 (make both automorphisms). If $\tilde{T}'$ is not yet surjective, apply Theorem 4.1 to $\tilde{T}'$ to obtain a further enlargement on which $\tilde{T}'$ becomes an automorphism, and extend $\tilde{T}$ by the same conjugation construction (Steps 2–8). Since $\tilde{T}$ is already an automorphism, the conjugated extension is bounded by the same UBP argument. The resulting pair consists of commuting symplectic automorphisms extending T and T' . $\square$

Remark 6.2. The same argument gives the result for any finite family of mutually commuting symplectic embeddings (extend one after another by conjugation).

7 Examples

Example 7.1. If T is the unilateral block shift and $T' = T^m$, Theorem 6.1 recovers the bilateral shift and its m -th power as commuting symplectic automorphisms.

Example 7.2. Let $A, B \in \text{Sp}(2d)$ with $AB = BA$, and let S be the unilateral shift on $\ell^2(\mathbb{N})$. Then $T = S \otimes A$ and $T' = S \otimes B$ are commuting symplectic embeddings on $\ell^2(\mathbb{N}) \otimes \mathbb{R}^{2d}$, not necessarily Hilbert-space isometries. Theorem 6.1 still applies.

Remark 7.3. In finite dimensions, $T^* \mathcal{J} T = \mathcal{J}$ forces T to be invertible, so no enlargement is needed. The phenomenon is infinite-dimensional.

References

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